COOLANT PERFORMANCE

The required ethylene-glycol (antifreeze) and water mixture depends upon climate and vehicle operating conditions. The coolant performance of various mixtures follows:

Pure Water- Water can absorb more heat than a mixture of water and ethylene-glycol. This is for purpose of heat transfer only. Water also freezes at a higher temperature and allows corrosion.

100 percent Ethylene-Glycol - The corrosion inhibiting additives in ethylene-glycol need the presence of water to dissolve. Without water, additives form deposits in system. These act as insulation causing temperature to rise to as high as 149°C (300°F). This temperature is hot enough to melt plastic and soften solder. The increased temperature can result in engine detonation. In addition, 100 percent ethylene-glycol freezes at -22°C (-8°F).

50/50 Ethylene-Glycol and Water - Is the recommended mixture, it provides protection against freezing to -37°C (-34°F). The antifreeze concentration **must always** be a minimum of 44 percent, year-round in all climates. If percentage is lower, engine parts may be eroded by cavitation. Maximum protection against freezing is provided with a 68 percent antifreeze concentration, which prevents freezing down to -67.7°C (-90°F). A higher percentage will freeze at a warmer temperature. Also, a higher percentage of antifreeze can cause the engine to overheat because specific heat of antifreeze is lower than that of water.

CAUTION: Richer antifreeze mixtures cannot be measured with normal field equipment and can cause problems associated with 100 percent ethylene-glycol.

COOLANT SELECTION AND ADDITIVES

The use of aluminum cylinder blocks, cylinder heads and water pumps requires special corrosion protection. Only Mopar® Antifreeze/Coolant, 5 Year/100,000 Mile Formula (glycol base coolant with corrosion inhibitors called HOAT, for Hybrid Organic Additive Technology) is recommended. This coolant offers the best engine cooling without corrosion when mixed with 50% distilled water to obtain to obtain a freeze point of -37°C (-35°F). If it loses color or becomes contaminated, drain, flush, and replace with fresh properly mixed coolant solution.

CAUTION: Do not use coolant additives that are claimed to improve engine cooling.

TRANSFER CASE - NV231

Recommended lubricant for the NV231 transfer case is Mopar® ATF +4, Automatic Transmission Fluid.

TRANSFER CASE - NV241

Recommended lubricant for the NV241 transfer case is Mopar® ATF +4, Automatic Transmission Fluid.

AXLE LUBRICATION

NOTE: DiamlerChrysler recommends using Mopar® lubricants or lubricants of equal quality.

FRONT AXLE

- 181 FBI (Model 30) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)
- RUBICON 216 FBI (Model 44) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)

REAR AXLE

- 194 RBI (Model 35) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)
- 194 RBI (Model 35) 4.56 Ratio - 2.4 L Enigne and 42 RLE Automatic Transmission - Mopar® Synthetic Gear Lubricant 75W-140
- 226 RBI (Model 44) - Mopar® Gear Lubricant 80W-90 (Trailer Towing Mopar® Synthetic Gear Lubricant 75W-140)
- RUBICON 226 RBI (Model 44) - Mopar® Synthetic Gear Lubricant 75W-140

NOTE: Trac-lok® equipped axles require 118 ml (4 ounces) of Limited Slip Additive in the lubricant.

MANUAL TRANSMISSION

NOTE: DaimlerChrysler recommends using Mopar® lubricants or lubricants of equal quality.

- NSG370 - Mopar® Manual Transmission Lubricant MS-9224

AUTOMATIC TRANSMISSION FLUID

NOTE: Refer to Service Procedures in this group for fluid level checking procedures.